

Interactive DAS dataset with paired and independent t-test results computed live

SAMPLE DATASET — 20 STUDENTS, SEGMENT C (AT-RISK REBUILDERS)

Student	Arm	Grade	Pre score	Post score	Delta	Dropped out	Geo index
S001	Control	8	62	60	-2	Yes	0.22
S002	Control	9	58	55	-3	No	0.31
S003	Control	8	71	69	-2	No	0.18
S004	Control	10	65	63	-2	Yes	0.28
S005	Control	9	54	52	-2	Yes	0.35
S006	Control	8	68	65	-3	No	0.24
S007	Control	10	60	58	-2	Yes	0.29
S008	Control	9	57	56	-1	No	0.20
S009	Control	8	73	71	-2	No	0.33
S010	Control	10	61	59	-2	Yes	0.27
S011	Treatment	9	63	68	+5	No	0.21
S012	Treatment	8	56	62	+6	No	0.30
S013	Treatment	10	70	77	+7	No	0.19
S014	Treatment	8	64	70	+6	No	0.26
S015	Treatment	9	55	61	+6	Yes	0.34
S016	Treatment	10	67	74	+7	No	0.23
S017	Treatment	8	59	65	+6	No	0.28
S018	Treatment	9	58	65	+7	No	0.22
S019	Treatment	10	72	79	+7	No	0.32
S020	Treatment	8	62	69	+7	No	0.25

TEST 1 — PAIRED T-TEST (WITHIN EACH ARM)

Control arm: did scores change from baseline?		Treatment arm: did scores change from baseline?	
Mean pre score	62.9	Mean pre score	62.6
Mean post score	60.8	Mean post score	69.0
Mean delta ($\bar{d}$)	-2.10	Mean delta ($\bar{d}$)	+6.40
Std dev of deltas	0.57	Std dev of deltas	0.70
t-statistic	-11.699	t-statistic	28.945
p-value (two-tailed)	< 0.001	p-value (two-tailed)	< 0.001
Significant decline		Significant improvement	

TEST 2 — INDEPENDENT T-TEST (TREATMENT DELTA VS CONTROL DELTA)

Did the treatment arm improve significantly more than control?

Mean delta — control	Mean delta — treatment	Difference
-2.10	+6.40	+8.50
Pooled std error		0.285
t-statistic		29.845
Degrees of freedom		17.3
p-value (two-tailed)		< 0.001
Cohen's d (effect size)		13.347 (large effect)
Treatment significantly outperforms control		

PYTHON CODE

```
import numpy as np
import pandas as pd
from scipy import stats

# — Sample dataset —————
data = {
    "student_id": ["S001", "S002", "S003", "S004", "S005",
                  "S006", "S007", "S008", "S009", "S010",
                  "S011", "S012", "S013", "S014", "S015",
                  "S016", "S017", "S018", "S019", "S020"],
    "arm":        ["control"]*10 + ["treatment"]*10,
    "grade":      [8,9,8,10,9,8,10,9,8,10,
                  9,8,10,8,9,10,8,9,10,8],
    "pre_score":  [62,58,71,65,54,68,60,57,73,61,
                  63,56,70,64,55,67,59,58,72,62],
    "post_score": [60,55,69,63,52,65,58,56,71,59,
                  68,62,77,70,61,74,65,65,79,69],
    "dropped_out": [1,0,0,1,1,0,1,0,0,1,
                   0,0,0,0,1,0,0,0,0,0],
    "geo_index":  [0.22,0.31,0.18,0.28,0.35,
                  0.24,0.29,0.20,0.33,0.27,
                  0.21,0.30,0.19,0.26,0.34,
                  0.23,0.28,0.22,0.32,0.25]
}

df = pd.DataFrame(data)
df["delta_score"] = df["post_score"] - df["pre_score"]

ctrl = df[df["arm"] == "control"]
trt = df[df["arm"] == "treatment"]

# — Test 1: Paired t-test (within each arm) —————
# H0: mean delta = 0 (no change from baseline)
t_ctrl, p_ctrl = stats.ttest_rel(ctrl["post_score"], ctrl["pre_score"])
t_trt, p_trt = stats.ttest_rel(trt["post_score"], trt["pre_score"])

print(f'Control — t={t_ctrl:.3f}, p={p_ctrl:.4f}')
```

```

print(f'Treatment — t={t_trt:.3f}, p={p_trt:.4f}')

# — Test 2: Independent t-test (delta: treatment vs control)
# H0: mean(delta_trt) == mean(delta_ctrl)
t_ind, p_ind = stats.ttest_ind(
    trt["delta_score"],
    ctrl["delta_score"],
    equal_var=False # Welch's — safer default
)

# Cohen's d
pool_std = np.sqrt((trt["delta_score"].std()**2 +
                    ctrl["delta_score"].std()**2) / 2)
cohens_d = (trt["delta_score"].mean() -
            ctrl["delta_score"].mean()) / pool_std

print(f'Independent — t={t_ind:.3f}, p={p_ind:.4f}, d={cohens_d:.3f}')

```

Why Welch vs standard? ↗

Interpreting Cohen's d ↗

Run ANCOVA instead ↗